

Agile Development of PressAI: Enhancing Educational Efficiency through QR-Based Attendance and AI-Powered Essay Scoring

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Abstract: In learning activities, common issues such as time-consuming and manipulation-prone attendance processes, as well as subjective and inconsistent essay grading, often arise. This study aims to develop PressAI, an AI-based application that automates attendance using QR code technology and provides objective essay grading powered by the OpenAI API. The application was developed using the Agile methodology integrated within the Software Development Life Cycle (SDLC) framework, through iterative stages including planning, requirements analysis, system design, implementation, and testing. The results demonstrate that PressAI enables instant and accurate attendance recording via QR scanning, provides a seamless login experience without repeated authentication, and delivers fast and objective essay assessments. With its features, PressAI is expected to enhance efficiency and transparency in the learning process while offering opportunities for further innovative feature development.

Kata Kunci: Attendance, Automatic Essay Scoring, Agile, Artificial Intelligence

I. INTRODUCTION

Attendance is an important activity that is routinely carried out in various aspects of society, including in offices, construction, and education [1]. Attendance serves as a key indicator for measuring productivity and discipline, particularly in the field of education [2]. However, traditional attendance methods often consume considerable time, thereby reducing the efficiency of the learning process [3]. Challenges like these have led to various technological innovations, one of which is the use of QR codes (Quick Response), developed by Denso Wave in Japan in 1994 [4]. This technology facilitates and speeds up the attendance process while reducing the potential for fraud, such as proxy attendance [5], [6].

QR technology has been widely tested and proven effective in various contexts. For instance, a study conducted at STIMIK AMIKBANDUNG highlighted that the use of a QR-based system offers several advantages, including minimizing fraud in the attendance process, effectively eliminating proxy attendance practices [7]. Another study mentioned additional benefits, such as the efficiency in recording attendance and the ease for educators in managing data, thus benefiting not only students but also teachers and educational institutions in collecting attendance data [8].

Apart from attendance issues, essay task evaluations are another common challenge in the education sector. Essay assessment often sparks debates due to its subjective nature, eliciting complaints from students who perceive the grading as unfair and from educators who struggle to maintain consistency and objectivity [9], [10].

To address this, solutions like Automated Essay Scoring (AES) have been developed, employing Natural Language Processing (NLP)-based approaches [11]. These technologies integrate deep learning models, such as BERT

(Bidirectional Encoder Representations from Transformers), and hybrid models to enhance scoring accuracy and provide more objective feedback [12]. However, these solutions still face limitations, such as difficulty capturing linguistic nuances, potential biases in training data, and constraints in offering in-depth evaluations [11], [13], [14].

As a more innovative solution, the use of APIs from OpenAI introduces a new approach that goes beyond conventional NLP. OpenAI leverages the power of Generative AI to not only process and evaluate text but also generate dynamic responses based on context [15]. This enables the development of smarter applications, such as more efficient attendance systems and essay evaluations that are not only swift and objective but also capable of providing personalized and in-depth feedback to students [16].

The advantage of using OpenAI APIs lies in their ability to comprehend language in a multi-dimensional manner, offer contextual predictions, and provide more flexible solutions compared to traditional AES approaches [17]. This enables educators to not only reduce their workload but also enhance the quality of education through more adaptive and transparent technology. Thus, the development of OpenAI API-based applications is expected to become a significant innovation in addressing the challenges of attendance and essay evaluation in the education sector.

II. METHOD

The research employs the Agile methodology as the foundation for its development process. Agile is a dynamic, iterative, and incremental approach to software development that prioritizes collaboration among cross-functional teams, adaptability, continuous refinement, and swift responses to evolving requirement [18]. This methodology divides the work into short cycles, known as "sprints," enabling teams to

regularly evaluate project progress and make necessary refinements. These refinements are documented as the sprint backlog, and for this study, the development has been structured into three planned sprints. By focusing on consistent delivery of functional features and fostering regular communication with stakeholders, Agile aims to enhance both customer satisfaction and development efficiency.

This study follows the Agile methodology within the framework of the Software Development Life Cycle (SDLC), encompassing several essential stages: planning, requirements analysis, system design, implementation, and testing, as shown in Figure 1 [19]. Agile was chosen as the development strategy for this software due to its ability to deliver results more rapidly, with improved quality and reliability, making it particularly suited to the dynamic and competitive landscape of mobile application development [20].

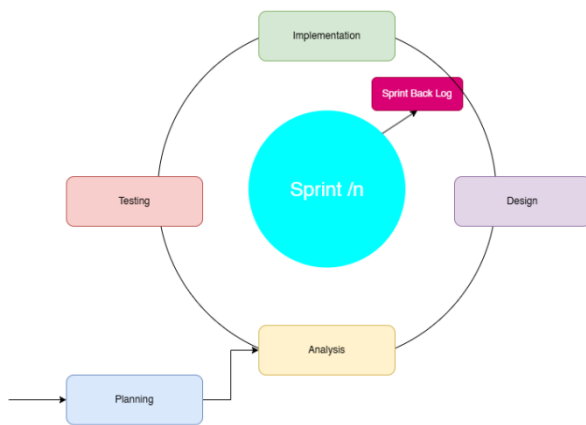


Figure 1. stages of the agile method

The development of the PressAI application using Agile followed these main phases, adapted from the Software Development Life Cycle (SDLC):

a. Planning

The project began by identifying the objectives and scope, which included creating an application for QR-based attendance and automated essay evaluation using AI. Key stakeholders, including developers, educators, and technical experts, collaborated to prioritize features and define a clear roadmap for development. Sprint planning sessions were held to outline tasks and goals for each sprint cycle.

b. Requirements Analysis

During this phase, the team gathered functional and non-functional requirements for the application through discussions with end-users and stakeholders. Essential functionalities, such as QR-based attendance, essay evaluation, and data management, were documented. Non-functional requirements like security, performance, and reliability were also identified to guide development.

c. System Design

The design phase involved creating user interface mockups, system architecture diagrams, and database

schemas. These designs served as blueprints to guide development. Prototypes of the application's core features, such as the QR scanning module and AI evaluation system, were iteratively reviewed and refined with input from stakeholders during sprint reviews.

d. Implementation

The development team implemented the planned features incrementally in sprint cycles. Each sprint focused on delivering functional components, such as the QR code scanning module, AI-based essay evaluation system, and user authentication. Regular stand-up meetings ensured progress tracking, and issues were addressed promptly.

e. Testing

Testing was conducted iteratively throughout the development process. Each sprint included unit testing, integration testing, and user acceptance testing for completed features. Black-box testing was employed to validate that all functionalities met the specified requirements. Feedback from stakeholders during sprint reviews was incorporated into subsequent iterations to improve the application's performance and usability.

III. RESULT AND DISCUSSION

After following the steps according to the procedure and aligning with the prepared plan for developing this application, we have achieved the following outcomes:

a. Planning

We collaborated with clients to gather initial system requirements and define the project's objectives. During this phase, a feasibility study was conducted to evaluate whether the application was viable for development. This assessment included technical and organizational analyses to ensure the project's feasibility and alignment with stakeholder expectations [21].

1. Technical Feasibility Study

In the technical feasibility study, we determined that the risks associated with familiarity in developing similar applications are manageable, as most team members have sufficient experience in creating similar projects. Furthermore, the risk related to familiarity with technology is highly manageable since all team members are proficient in using the required technology. However, there are challenges regarding the project's scale, as the team consists of only four members, with three actively involved in project development. The estimated timeframe of three months poses potential risks, though the team is committed to ensuring the project is completed within the deadline. Finally, system compatibility poses a relatively low risk, as the application is built using Java, which supports Android systems but currently lacks compatibility with iOS and desktop platforms.

2. Organizational Feasibility Study

The organizational feasibility risks are minimal because the purpose of developing this application is to enhance efficiency, accuracy, and engagement in the processes of attendance and essay evaluation within academic

environments. The project involves collaboration with various stakeholders, including a faculty member from the Department of Informatics and Computer Engineering Education to ensure the application meets educational needs and aligns with academic standards. Involving stakeholders such as faculty members ensures that the application addresses practical challenges in educational settings while fostering active participation and support from the academic community. This collaboration strengthens the organizational foundation for the successful implementation of the application.

b. Analysis

The analysis phase involved identifying functional requirements such as user registration, QR-based attendance, automated essay evaluation, and data management, along

with non-functional requirements like performance, security, and reliability. Additionally, a use case diagram outlined user interactions with the system, and activity diagrams detailed workflows for login, attendance scanning, and essay evaluation processes.

1. Functional Requirements

Functional requirements refer to the features directly used by users to operate the application, without which the application cannot function. In addition to addressing user needs, functional requirements also serve to analyze the necessary elements for the application being developed [22]. Functional requirements facilitate the development process, making it more streamlined. For further clarity, refer to Table 1 below.:

Table 1. Functional Requirement	
Requirement	Description
User Registration	Enables users to register and create user accounts.
Authentication	Provides authentication to ensure that only registered users can access the application.
QR Code Attendance	Allows lecturers to generate unique QR codes for each session, and students to scan these codes to record their attendance.
Automated Essay Grading	Enables students to submit their essay answers via the application and uses the GPT API to evaluate and assign automatic grades based on a predefined answer key.
Student Data Management	Allows lecturers to view and manage student data, including attendance records and essay grades.
Attendance History	Provides an attendance history for each student, including dates and times of attendance.
Test Result History	Provides a history of test results for each student, including scores from all tests taken. This helps students and lecturers track academic progress over time.
API Integration	Integrates APIs to process QR codes for attendance and evaluate essay answers.

2. Non-Functional Requirements

Non-functional requirements, in contrast to functional requirements, refer to the features of the application that are not directly used by users but operate in the background [23]. Despite running in the background, these requirements

remain critical, such as application security, which is essential for protecting user privacy [24]. Non-functional requirements play a crucial role in refining the application and ensuring its overall quality. For further clarity, refer to Table 2 below :

Table 2. Non Functional Requirement	
Requirement	Description
Performance	The application must be responsive and capable of accommodating a large number of users without significant performance degradation.
Reliability	The application must operate without significant disruptions or failures and should quickly recover in case of failure.
Security	The application must have a high level of security to protect sensitive user data, such as attendance and grades, from unauthorized access.
Availability	The application must be available to users at all times, except during scheduled maintenance or updates.
Usability	The user interface must be intuitive and easy to use, allowing users to interact with the application with ease.

Portability	The application must run on various platforms and hardware environments without significant modifications.
Documentation Availability	Complete documentation must be provided to help users effectively use the application.
Resource Efficiency	The application must use system resources (such as CPU, memory, and network) efficiently to optimize performance and resource usage.

3. Use Case Diagram

The use case describes how the application will function and interact with users to achieve specific goals and tasks [25], and the details can be seen in Figure 2 :

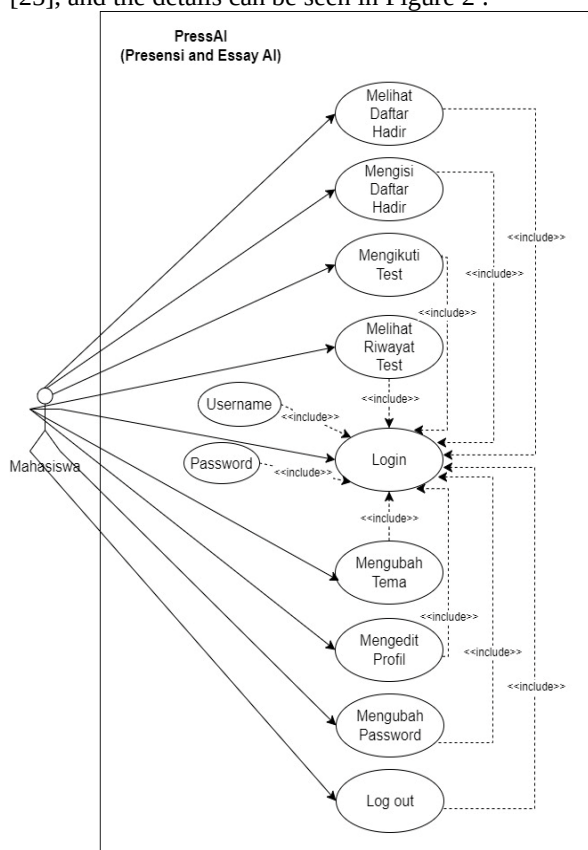


Figure 2, Use Case Diagram

In Figure 2, we can see that the use case in our application consists of only one actor, which is the student. In this case, the student can log in, mark attendance, and take tests, after which the data is sent to the database for processing by both the AI and the lecturer, each using a different platform, namely the website. This is why in our application, we only list the student as the actor.

4. Activity Diagram

The activity diagram illustrates the flow of activities within our application, showing how each process, as outlined in the use case, will be executed [26]. In our application, as described in the use case, there are three main activities: Login, Attendance, and Essay Grading. Below is the activity diagram for each of these activities

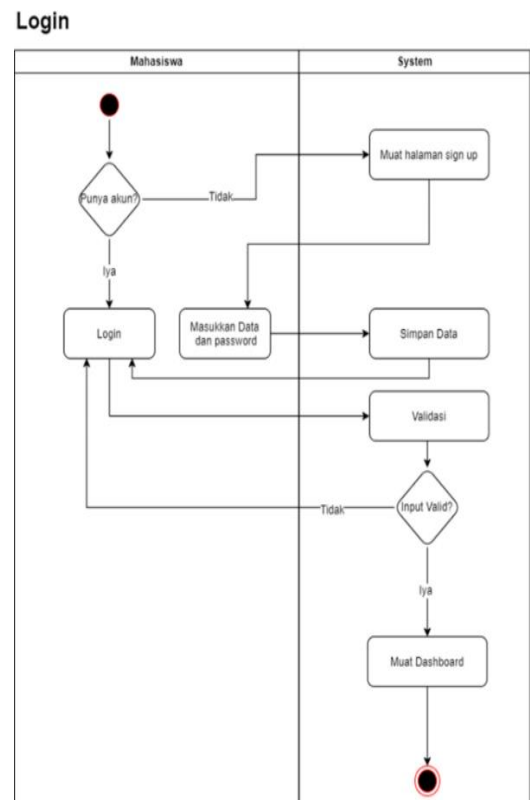


Figure 3. Login Activity

As we can see in figure 3, the login activity diagram show that the process starts with verifying whether an account exists. If no account is found, the user will be directed to sign up first. After account creation, the user can log in, and the input will be validated. If the input is correct, the user will be directed to the dashboard. However, if the input is incorrect, a warning will be displayed.

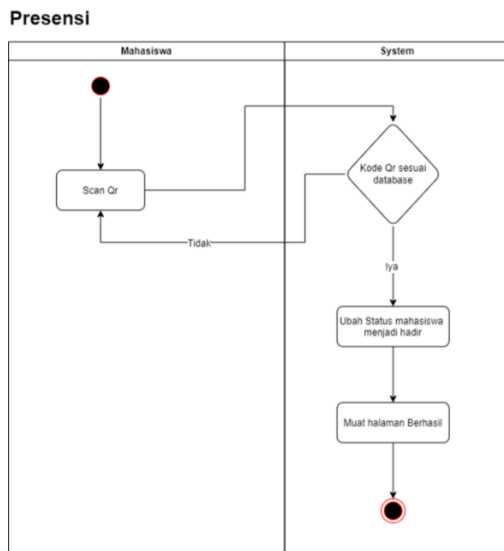


Figure 4. Attendance Activity

Next, in Figure 4, we can see the activity that occurs in the Attendance section. The attendance will be conducted using a QR scan, and the user's data will be automatically saved into the database. If the correct code is scanned, the attendance history will be saved. However, if an incorrect code is scanned, nothing will happen.

Penilaian Essay

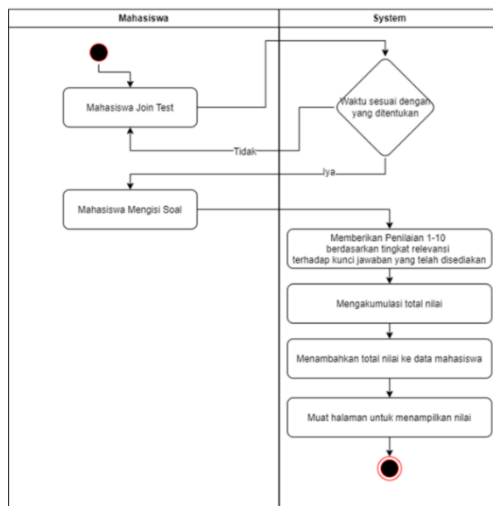


Figure 5. Essay Assesment Activity

Finally, in Figure 5, the activity of Essay Grading is explained. It starts with the student scanning a specific QR code for the essay test. If the correct QR code is scanned, the student will be directed to the waiting lobby to begin the test. However, if an incorrect code is scanned, nothing will happen. After waiting for the designated time, the student can proceed to answer the essay questions. Once the student has completed the test, the score will automatically be generated using AI.

c. Design

After the analysis phase, we move on to the design phase, where we will provide an overview of the design of our application, starting with the UI (User Interface), followed by the ERD (Entity-Relationship Diagram), class diagram, and finally the sequence diagram. Below are the explanations:

1. User Interface

The user interface represents the view that users will see in our application. The UI is designed to ensure that users can quickly and easily achieve their goals. This includes clear navigation, easily accessible buttons, and a logical user flow for maximum efficiency. Additionally, the application focuses on providing a positive and engaging experience to enhance user satisfaction [27]. For our UI design, we used Figma to create interactive prototypes that allow for an intuitive and seamless user experience. Below is the UI design of our application:

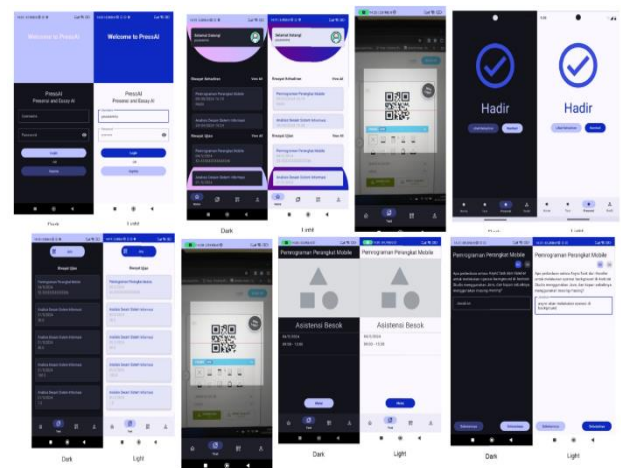
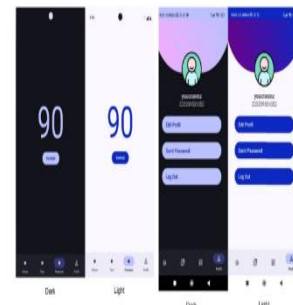


Figure 6. UI Design

As shown in Figure 6, the design of our application has been developed, starting from the login page to the dashboard. This design focuses on providing a seamless user experience, ensuring that the interface is intuitive and easy to navigate for users.

2. Entity-Relationship Diagram



Figure 7. Entity-Relationship Diagram

The Entity-Relationship Diagram is a diagram used to illustrate the relationships between entities in a database [28]. With this diagram, it becomes easier for us to develop our application, as shown in Figure 7. In our ERD, there are 9 entities, each of which has a Primary Key or a Foreign Key, serving different functions. The Primary Key is an attribute or set of attributes in a table that uniquely identifies each row in that table, while the Foreign Key is used to establish relationships between tables in a relational database.

3. Class Diagram

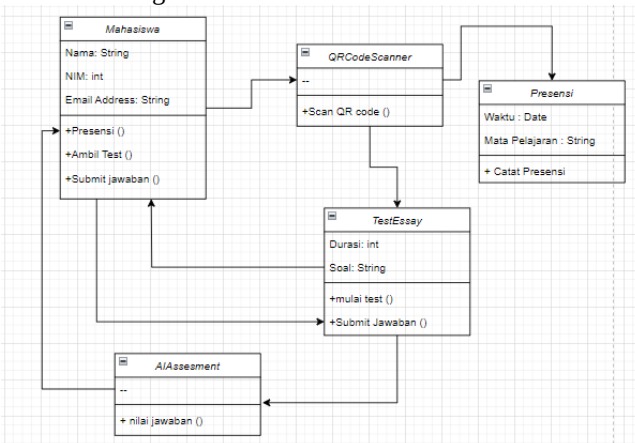


Figure 8. Class Diagram

In Figure 8, we can see the Class Diagram, which is part of the Unified Modeling Language (UML) used to represent the static structure of a software system. It shows the classes, attributes, methods, and relationships between classes [29]. Below is the relationship between the 5 classes above:

1. Student: The student has access to the QRCodeScanner to interact with the Attendance and TestEssay classes. After connecting, the

student can access the TestEssay class by submitting their answers.

2. QRCodeScanner: This class serves only as an intermediary, connecting the student to the TestEssay and Attendance classes.
3. TestEssay: This class has access to the student after providing the questions and also receives the student's answers, which are then sent to AiAssessment for grading.
4. Presensi (Attendance): Connected to the student through the QRCodeScanner, it has limited access compared to other classes, as its primary function is to store the student's attendance data.
5. AiAssessment: This class has access to the student to directly send the grading results to them.

4. Sequence Diagram

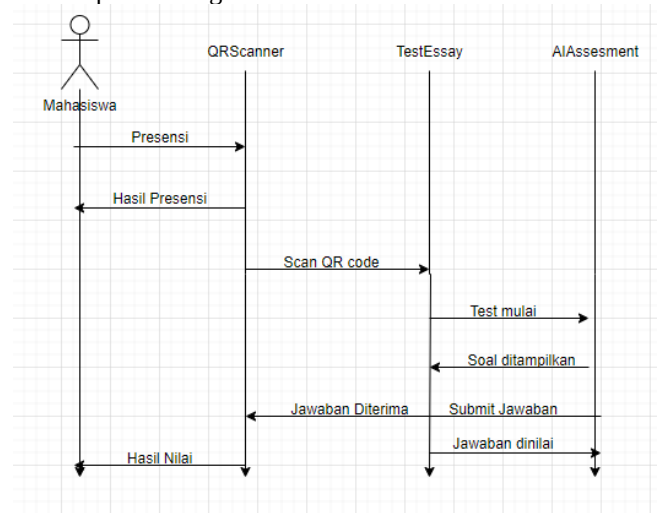


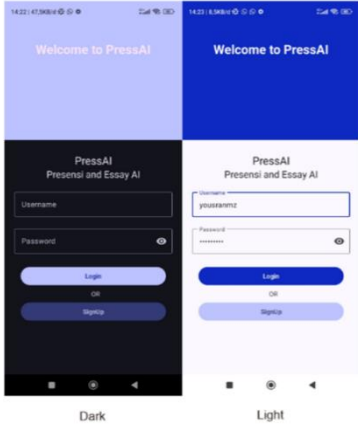
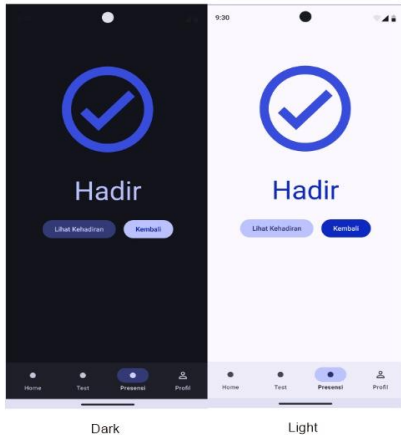
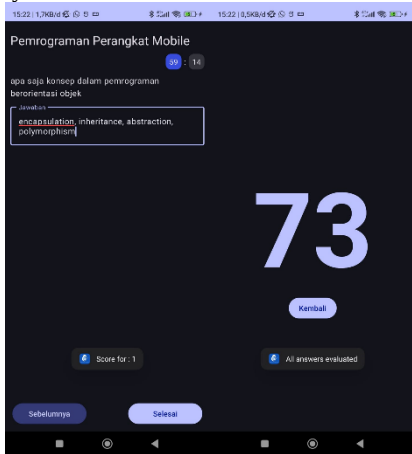
Figure 9. Sequence Diagram

The Sequence Diagram is a tool used to show the steps taken by the computer in a specific order. It helps us visualize how the system will execute each command sequentially, as seen in Figure 9 [30]. The process starts with attendance, then proceeds to the test, and ends with providing the results. This process is carried out in sequence according to the procedures that have been planned in advance.

d. Implementation

In the implementation phase, the application is developed using Java as the primary programming language, leveraging Android Studio as the integrated development environment (IDE) for building the mobile application. Firebase is used as the database to manage user data, attendance records, and test results, ensuring real-time synchronization and reliable data storage. For the AI-driven essay grading feature, the application integrates OpenAI's API to process and evaluate student submissions automatically. This combination of technologies enables the creation of a responsive, secure, and efficient application tailored to the needs of students and educators.

Table 3. Development Result

Coding	Result
Login <pre>11 @Override 12 protected void onCreate(Bundle savedInstanceState) { 13 SharedPreferences sharedPref = getSharedPreferences("name: 'MyPrefs'", Context.MODE_PRIVATE); 14 String username = sharedPref.getString("key: 'username'", null); 15 super.onCreate(savedInstanceState); 16 17 if (username == null) { 18 Intent dashboardIntent = new Intent(getApplicationContext(), Dashboard.class); 19 startActivity(dashboardIntent); 20 finish(); 21 } else { 22 startActivity(R.layout.activity_main); 23 24 login_btn = findViewById(R.id.login_btn); 25 login_signup_btn = findViewById(R.id.login_signup_btn); 26 input_username = findViewById(R.id.input_username); 27 input_password = findViewById(R.id.input_password); 28 edit_text_username = findViewById(R.id.edit_text_username); 29 edit_text_password = findViewById(R.id.edit_text_password); 30 31 login_btn.setOnClickListener(new View.OnClickListener() { 32 @Override 33 public void onClick(View v) { 34 String username = input_text_username.getText().toString(); 35 36 // ... (rest of the code) 37 } 38 }); 39 } 40 }</pre>	Login 
Attendance <pre>11 @Override 12 public void onCreate(Bundle savedInstanceState) { 13 Bundle savedInstanceState; 14 View view = inflater.inflate(R.layout.fragment_presensi, container, false); 15 barcodeView = view.findViewById(R.id.qr_scanner); 16 barcodeView.setStatusText(""); 17 SharedPreferences sharedPref = getActivity().getSharedPreferences("name: 'MyPrefs'", Context.MODE_PRIVATE); 18 if (ContextCompat.checkSelfPermission(getActivity(), Manifest.permission.CAMERA) != PackageManager.PERMISSION_GRANTED) { 19 ActivityCompat.requestPermissions(getActivity(), new String[]{Manifest.permission.CAMERA}, REQUEST_CODE); 20 } else { 21 @Override 22 barcodeView.decodeSingle(new BarcodeCallback() { 23 @Override 24 public void decodeResult(BarcodeResult result) { 25 String barcodeText = result.getText(); 26 DatabaseReference ref = FirebaseDatabase.getInstance().getReference().child("siswa_kuliah"); 27 28 @Override 29 ref.addListenerForSingleValueEvent(new ValueEventListener() { 30 @Override 31 public void onDataChange(DataSnapshot dataSnapshot) { 32 for (DataSnapshot childSnapshot : dataSnapshot.getChildren()) { 33 34 // ... (rest of the code) 35 } 36 } 37 }); 38 } 39 }); 40 } 41 }</pre>	Attendance 
Test Essay <pre>11 @Override 12 public void onCreate(Bundle savedInstanceState) { 13 super.onCreate(savedInstanceState); 14 if (getArguments() != null) { 15 wParam = getArguments().getString(ARG_PARAM1); 16 wParam2 = getArguments().getString(ARG_PARAM2); 17 } 18 19 @Override 20 public void onCreate(Bundle savedInstanceState) { 21 // Inflate the layout for this fragment 22 View view = inflater.inflate(R.layout.fragment_test, container, false); 23 barcodeView = view.findViewById(R.id.qr_scanner); 24 barcodeView.setStatusText(""); 25 if (ContextCompat.checkSelfPermission(getActivity(), Manifest.permission.CAMERA) != PackageManager.PERMISSION_GRANTED) { 26 ActivityCompat.requestPermissions(getActivity(), new String[]{Manifest.permission.CAMERA}, REQUEST_CODE); 27 } 28 29 @Override 30 barcodeView.decodeSingle(new BarcodeCallback() { 31 @Override 32 public void decodeResult(BarcodeResult result) { 33 34 // ... (rest of the code) 35 } 36 }); 37 } 38 }</pre>	Test Essay 

Link Github: <https://github.com/Yousran/PressAI.git>

e. Testing
The testing method applied by the author to evaluate this application is black-box testing, which aims to check

whether the system requirements have been met, particularly in terms of functional specifications [31].

Table 4. Black-Box Testing Result

Test Case Number	System Tested	Expected Outcome	Test Results
TC1	Sign Up	If the input data (username, NIM, email, password, confirm password) is correct, the user will be directed to the login page, and the participant's data will be added to the database	✓ Valid
TC1	Login	If the username and password are correct, the user will be directed to the dashboard	✓ Valid
TC1	Attendance History	If the attendance history is present in the database, it will be displayed on the dashboard and Attendance tab	✓ Valid
TC1	Exam History	If the exam history is in the database, it will be displayed on the dashboard and Exam tab	✓ Valid
TC1	Scan Exam Code	If the exam code scanned matches the one in the database, the user will be directed to the exam lobby and the exam data will be shown	✓ Valid
TC1	Scan Attendance Code	If the scanned code matches the one in the database, a "successfully attended" message will be shown	✓ Valid
TC1	Display Exam Questions	When the start button on the exam lobby screen is pressed, the exam questions will be displayed	✓ Valid
TC1	Evaluate Exam Questions	When the finish button is pressed, the answers will be evaluated, and the total score will be displayed on the results page	✓ Valid
TC1	Automatically End Exam	If the timer reaches the exam duration, the exam will end and the answers will be evaluated	✓ Valid
TC1	Edit Profile	If the profile data is updated, the changes will be reflected in the database and displayed in the user interface	✓ Valid
TC1	Log Out	If the Log Out button is pressed, the user will be redirected to the login page	✓ Valid

In this study, the implementation of the application was carried out in three sprints, each focusing on different core features of the system. Initially, users were required to authenticate themselves using their username and password. However, following a stakeholder request, the system was enhanced to ensure that once users logged in, they would be redirected directly to the dashboard without the need for repeated authentication. This design mirrors concepts from prior research, such as in a cashier system, where login credentials are stored and users only need to log in once [32].

The second sprint focused on the attendance feature, which involved the integration of a barcode scanner to validate users' presence during specific class sessions. By scanning a QR code associated with the session, users' attendance was automatically recorded in the system. This feature is similar to methods used in previous studies, where QR codes were utilized for automatic attendance tracking. However, the key difference in this research was the application of the feature within an internship setting, where students were able to scan QR codes to confirm their attendance [33].

In the final sprint, the essay testing feature was developed. This feature enabled users to scan a QR code related to an essay test, where they would be redirected to the test lobby to view the details and start the test. The integration of OpenAI's API allowed the system to automatically evaluate users' responses. The JSON data sent to the API contained key elements such as the API key, model, prompt, question, answer, and context, with the response returning a numerical score that was compared to the correct answer. This feature enabled automated grading and provided immediate feedback to the users.

By dividing the project into these three sprints, we ensured a systematic and iterative approach to development, where each feature was implemented and tested individually. The division allowed for a more manageable workflow and provided opportunities for continuous feedback and adjustments as the project progressed. This method ensured that the system's core functionalities were built upon each other in a structured and efficient manner, resulting in a complete application that met the needs of both users and stakeholders.

IV. CONCLUSION

The Press AI application is expected to effectively support learning processes, particularly in internship environments, by providing practical solutions through its main features: login, attendance, and essay testing. These features were developed iteratively using the Agile method. The login feature offers users a seamless experience without repeated authentication, while the attendance feature utilizes QR code technology for automatic attendance recording. The essay testing feature integrates OpenAI's API to enable automated evaluation of responses. With its innovative design, Press AI has the potential to enhance accessibility, organization, and functionality in educational settings.

For future research, it is recommended to explore ways to enhance the functionality and accessibility of the system. Features such as real-time notifications for students and instructors, integration with calendar systems for improved scheduling, and tools to generate detailed analytical reports on attendance and performance could significantly improve usability. Additionally, incorporating multilingual support and offline capabilities could expand the system's accessibility, ensuring it remains functional and reliable in diverse environments and scenarios.

Research could also focus on evaluating user acceptance through established frameworks such as the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) [34], [35]. TAM could be utilized to analyze how enhancements in perceived usefulness and ease of use, driven by intuitive interfaces and seamless integration with educational tools, influence user adoption. UTAUT can provide insights into the impact of factors such as user training, support mechanisms, and social influence on the adoption process. Furthermore, implementing continuous feedback loops and regular updates aligned with user needs could improve trust and satisfaction, ensuring long-term acceptance and effectiveness of the system.

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